Ogbonna Arinze

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SUMMARY

Software Engineer with 8 years of professional experience building scalable backend services, REST APIs, and full-stack applications across fintech, healthcare, civic tech, and agency environments. Strong proficiency in Node.js, TypeScript, and PostgreSQL, with hands-on experience designing event-driven architectures, real-time streaming systems, and production API platforms serving web and mobile clients. Proven ability to deliver high-quality software in distributed, remote-first teams while collaborating with PMs, designers, and cross-functional stakeholders.

CORE COMPETENCIES

- **Backend & APIs:** Node.js, NestJS, REST API design, JWT authentication, RBAC, SSE streaming, event-driven architecture, and webhook integrations.
- **Programming Languages:** TypeScript, modern JavaScript (ESNext), Node.js, Python, and PHP.
- **Databases & Data:** PostgreSQL (pgvector), MongoDB, Prisma ORM, SQL query optimization, and caching patterns.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, Docker Compose, GitHub Actions, CI/CD pipelines, Traefik, Heroku, and Nx monorepo orchestration.
- **Frontend:** React, Next.js, Vue.js, Redux, Tailwind CSS, Radix UI, and responsive component systems.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Mastra, vector search (pgvector, Chroma), embeddings, and multi-agent workflows.
- **Leadership:** Code reviews, architecture discussions, mentorship, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, building both the API layer and conversational interface for real-time data querying.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, managing secure token handling via HTTP-only cookies and permission-based rendering across the application.
- Spearheaded the redesign of the main clinical trial data product, migrating from **Vue.js** to a **React Chrome extension**, optimizing workflows for clinical site integrations with upstream EHR systems.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** with **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Optimized application performance by implementing lazy loading, code splitting, dynamic imports, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, processing thousands to millions of daily transactions and boosting settlement rates by **80%** for MasterCard, VISA, Express-cash, and E-commerce modules.
- Adopted the native **EventSource API** for **real-time data streaming**, replacing conventional socket.io methods for faster updates, lower latency, and a more responsive interface for transaction monitoring.
- Developed a real-time account reconciliation solution, boosting daily reconciliation throughput by **60%** through innovative transaction processing and data management strategies.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, focusing on testable, maintainable, and well-documented code, while translating business requirements into efficient **MongoDB** database models and schemas.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**, reducing manual deployment overhead.

- Transformed Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface across the application.
- Mentored two interns through frontend and backend projects, providing technical guidance and structured task management.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications including user management, payroll processing, and attendance tracking.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, **NestJS** document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including **pgvector** extension for vector storage, and wrote custom migration tooling that filters out Mastra-managed chunk tables from Prisma's drift detection.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing **700+ documents** (959 files) across 37 Nigerian states with zero ingestion errors.
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats every 15s to prevent reverse proxy timeouts, token-budgeted context management, conversation summarization, and memory extraction.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a hybrid RAG pipeline combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests (4 projects), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*